

EBR Project Case Study

Executive Business Review | Global Electronics Retail

TOOLS USED: Excel Power Query | Pivot Tables | Pivot Charts | Excel Formulas | Camera Tool

\$29.2M	34%	\$18.3M	\$9.3M
US REVENUE	COMPUTERS CATEGORY SHARE	PEAK REVENUE (2019)	2020 REVENUE

The Business Problem

Global Electronics Retail needed a consolidated view of its commercial performance across 2016-2020. The objective was to help management understand changes in revenue and profitability, identify high- and low-performing products and categories, evaluate customer value, and monitor operational performance across markets.

The challenge was turning transactional sales data from multiple sources and currencies into a management-ready reporting system that could support both high-level executive review and deeper commercial analysis.

The Approach

I built an end-to-end Executive Business Review in Excel, combining Power Query, Pivot Tables, Pivot Charts, and formula-based calculations.

1. Data Preparation and Transformation

Power Query was used to prepare the underlying datasets and create a consistent analytical structure. Key transformations included:

- Cleaning and standardising the source data
- Preparing the sales data for analysis
- Merging the Exchange Rates table into the Sales table
- Converting transaction values into USD directly in Power Query
- Creating analysis-ready fields for reporting and segmentation

This ensured that the dashboard calculations were based on a consistent reporting currency rather than mixing values from different markets.

2. KPI Calculation

I created a dedicated calculation sheet containing the core business metrics using Excel formulas and Pivot-based measures. The executive overview tracks:

- Revenue
- Profit

- Profit Margin
- Orders
- Customers
- Units

The dashboard also incorporates year-over-year performance comparisons to make changes in business performance immediately visible.

3. Business Analysis

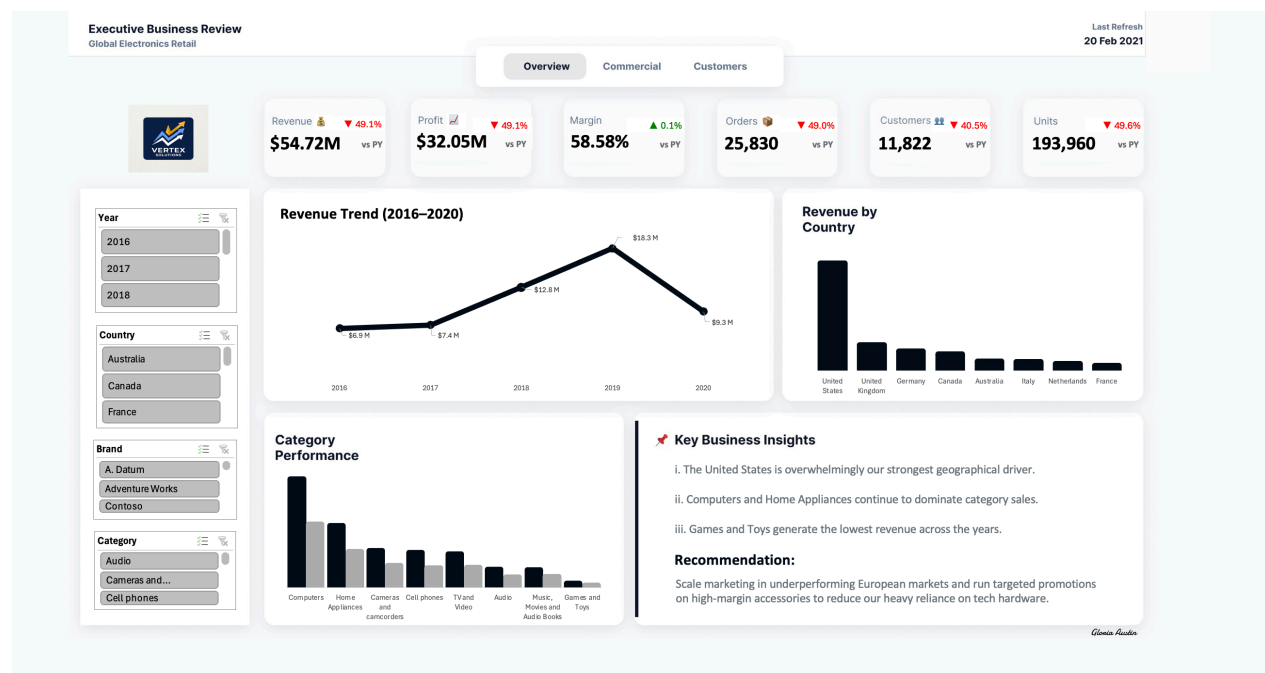
Pivot Tables and Pivot Charts were then used to analyse performance across several dimensions:

- Revenue trends by year
- Revenue by country
- Category performance
- Product performance
- Revenue vs. profit
- Category revenue contribution
- Customer value tiers
- Customer demographics
- Operational performance
- Regional performance

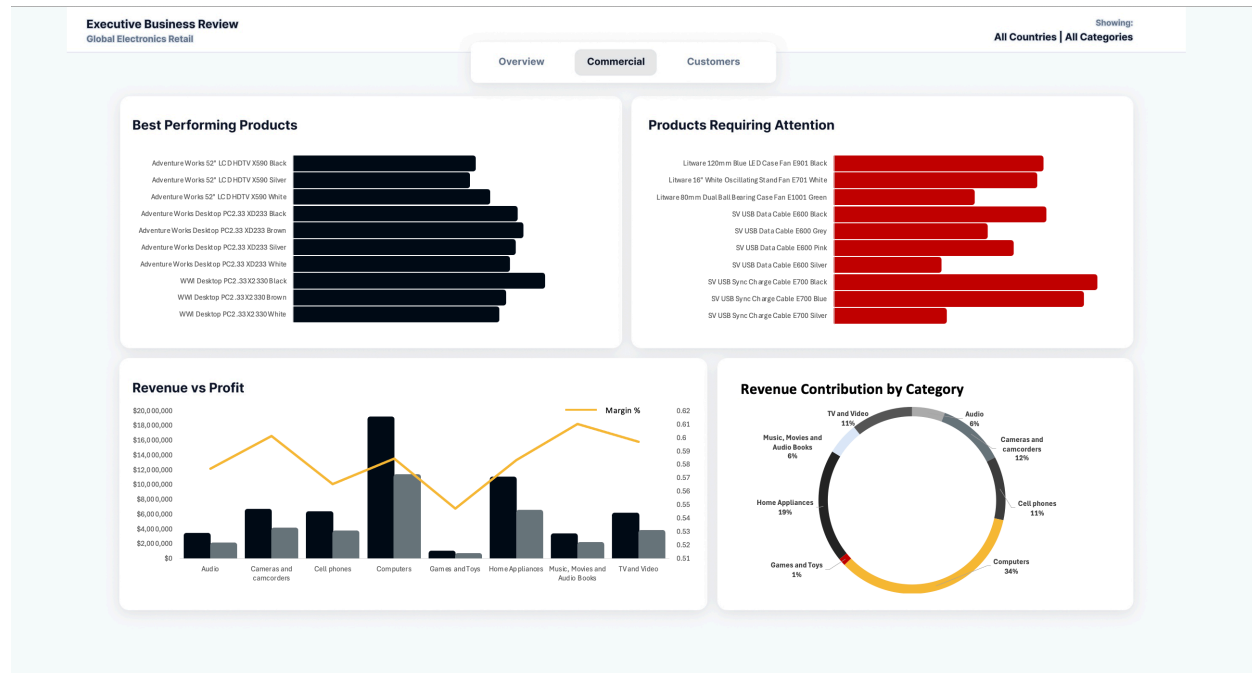
For the YoY analysis, I also used Excel's Camera Tool to integrate the comparison visuals into the executive dashboard without disrupting the dashboard layout.

Dashboard Design

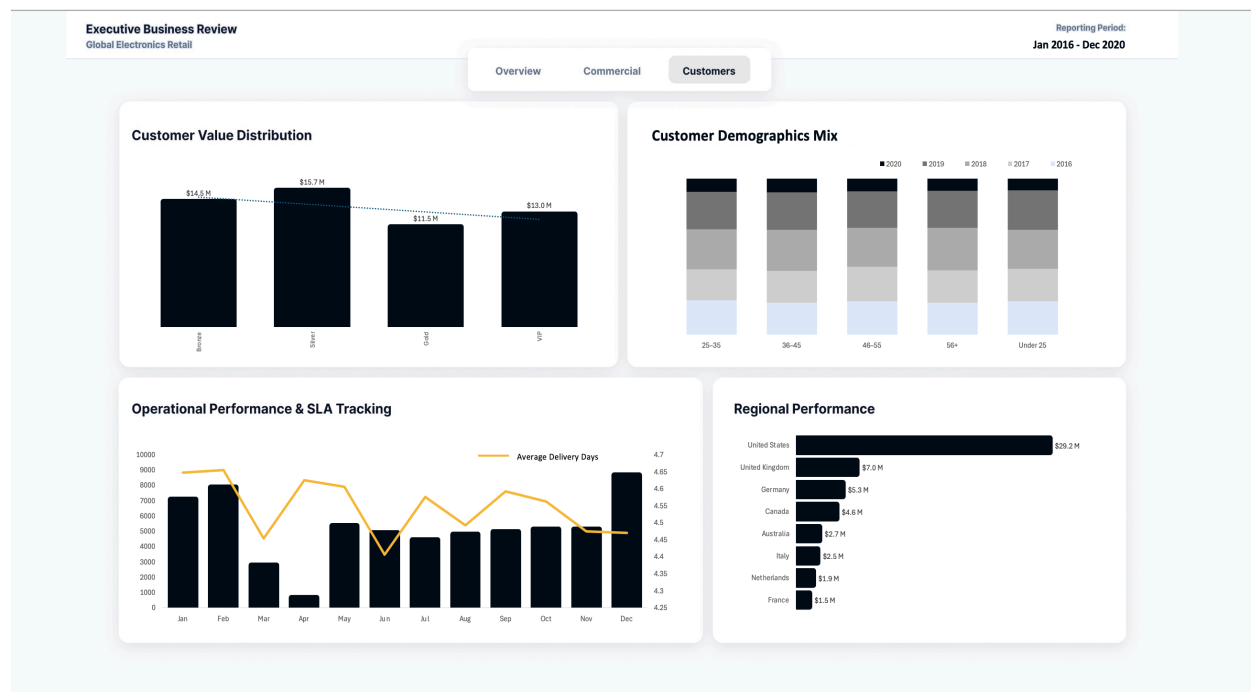
The final workbook was structured as three connected executive views:



- **Overview:** Provides management with the overall health of the business, combining KPI cards, revenue trends, geographical performance, category performance, and key business insights.



- **Commercial:** Focuses on product and category performance, highlighting both the strongest products and products requiring attention. It also compares revenue against profit and shows category contribution.



- **Customers:** Examines customer value, demographic composition, operational performance, and regional revenue contribution.

This structure allows an executive to move seamlessly from "How is the business performing?" to "What is driving performance?" to "Who and where are our customers?"

Key Findings

The analysis revealed several important commercial patterns:

- The United States was the strongest geographical revenue driver, contributing approximately \$29.2M.
- Computers represented the largest category contribution, accounting for approximately 34% of revenue.
- Home Appliances was the second-largest category contributor at approximately 19%.
- Games and Toys contributed only about 1%, making it the weakest category by revenue contribution.
- Revenue peaked in 2019 at approximately \$18.3M before falling to approximately \$9.3M in 2020.
- Several products emerged as clear high performers, while a separate group of products required commercial attention.
- Customer value was concentrated across the Silver, Bronze, VIP, and Gold segments, with Silver showing the highest value in the displayed analysis.
- Regional performance was heavily concentrated in the United States, with a substantial gap between the US and the remaining markets.
- Operational performance showed monthly variation in order volumes and average delivery time, providing an additional dimension for management monitoring.

Business Recommendation

The analysis suggests that management should reduce reliance on the strongest market and hardware-led categories by developing underperforming markets and strengthening higher-margin opportunities. In particular, the business could:

- Increase targeted marketing activity in weaker European markets.
- Investigate the causes of poor performance among products flagged for attention.
- Use promotions and cross-selling to increase contribution from weaker categories.
- Protect the strong performance of Computers and Home Appliances while diversifying the revenue mix.
- Monitor delivery performance alongside commercial KPIs to identify potential operational constraints.